BSc Project

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Characterisation of photon BEC phase diagram using machine learning

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Declaration of work undertaken

I hereby declare that this thesis was solely carried out by my project partner and I and that to the best of my knowledge, contains completely original content except for cases in which acknowledgement has been mentioned. The work was distributed evenly between us to maximise our efficiency as a team.

Abstract

A photon Bose-Einstein condensate (BEC) was classified using deep learning neural networks to produce two phase plots (for the two chosen classifications used) as functions of the input parameters to the system. The occupancy of the first 15 modes of the BEC were measured for a range of input parameters and were classified to be condensed based on whether their occupancy was greater than a threshold value. The first of the two classifications denoted the phase of the system by the total number of condensed modes of the first 15 - called the "length" classification. The other signified each set of 15 modes as a 15-bit binary number with each bit denoting whether each mode was condensed, with a 1 indicating condensed and vice versa - called the "binary" classification. The number of neurons and layers in two neural networks was optimised for each of the classifications, such that the accuracy of each neural network was optimised and so phase plots could be plotted. Given the input parameters, the length neural network predicted the total number of the first 15 modes which were condensed with uncertainty ± 1.21 and the binary neural network predicted the phase of the photon BEC exactly, 90.49% of the time.

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Chapter 1

Introduction

In the mid-1920s, Albert Einstein built upon work he received from Satyendra Nath Bose and predicted a new state of matter - a Bose-Einstein condensate (BEC) [1]. This was a system in which a large number of bosons (particles with integer spin) occupy the ground state such that the individual wavefunctions of the bosons superimpose. This causes quantum phenomena to become apparent on the macroscopic scale - something not seen before in matter.

In June 1995, a gaseous Rubidium condensate was produced by Eric Cornell and Carl Wieman [2], this was groundbreaking as it was the first of its kind, verified Einstein's hypothesis and eventually went on to win the Nobel prize.

In 2010 there was a further advancement in the field as the first photon BEC was produced [3]. This was achieved not by cooling as with atomic systems, but by pumping photons into a dye solution in an optical cavity at room temperature.

BEC research is very important due to the potential applications of the field. Some interesting examples include the inability of bunching qubits in quantum computers, decreasing quantum decoherence and making the new range of sensitive detectors (see to Appendix A).

Our project looks to the phase classification of a photon BEC using machine learning neural networks. This will allow the production of a phase diagram as a function of input parameters to the system.

We commence this project by initially, studying the theory behind photon BEC in Chapter 2.1 and neural network theory in Chapter 2.2. Chapter 3 moves on to how the method of classification of the energy modes of the system was achieved and how the neural networks used were optimised. Chapter 4 looks at the results of the project and to quantifying what they mean. Finally, Chapter 5 gives an overview of what was achieved in the project and some potential extensions which could improve the results obtained.

Chapter 2

Theory

2.1 Photon BEC

When discussing a Bose-Einstein condensate (BEC), it is often that we consider an atomic BEC – where bosons are cooled to extreme temperatures. This cooling causes the bosons to occupy the ground state due to their lack of energy.

To create a photon BEC, photons cannot simply be cooled as they have no chemical potential and so the system would just darken. Instead, at room temperature, a liquid dye is placed in an optical microcavity (a pair of close parallel mirrors, where at least one of which is curved) and pumped with photons using a photon pumping laser, Figure 2.1.

As a result of the geometry of the mirrors, the system has two potentials - a radial potential in the transverse plane (the plane parallel to the mirrors) and an infinite potential well in the longitudinal plane (the plane perpendicular to the mirrors). This gives the photons an effective mass and means they can be regarded as quasi-particles - making the system similar to a traditional atomic gas [2].

The curved mirrors used in photon BEC experiments are designed such the radial potential is a 2-dimensional quantum harmonic oscillator (QHO) (refer to Appendix B). The longitudinal potential well is created by the separation of the two mirrors. This potential has photons which occupy its energy modes with wavelengths proportional to the separation of the mirrors over an integer m. However, only one of its energy modes are occupied because the dye molecules (which absorb/emit the photons in the cavity) have the highest rates of absorption and emission in a small range of wavelengths - see Figure 2.2. This small range means that only the corresponding mode is occupied in the longitudinal potential, with a wavelength which we denote the cut-off wavelength, λ_0, which is also the wavelength which occupies the ground state of the radial potential. This cut-off wavelength is thus a function of the separation of the mirrors and can be changed to change the cut-off wavelength of the system (a parameter in changing the phase of the photon BEC).

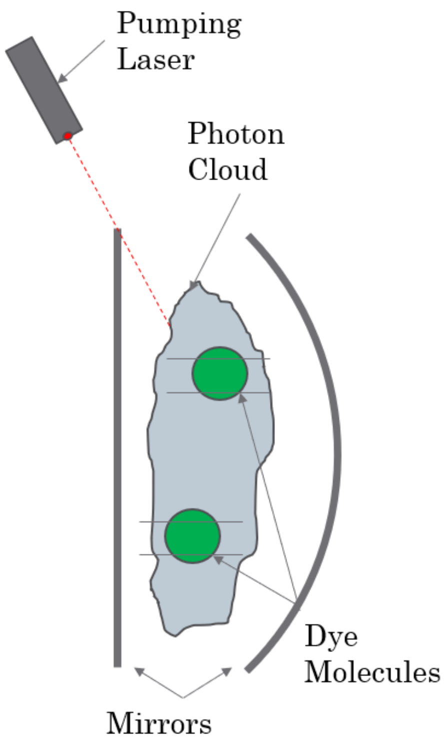


Figure 2.1: Set up of the photon BEC experiment. It consists of a pumping laser, pumping photons at a dye solution in an optical microcavity - composed of two mirrors. The dye molecules have discrete energy levels with energy gaps with which photons are absorbed and emitted with energy equal to (absorption and emission occurs through the exciting and de-exciting of the electrons of the dye molecules). The occupancy of each of the modes is measured using a charge-coupled device (CCD) and interferometric techniques.

The effective mass of the photons means that they also have an effective chemical potential, μ [11]:

$$\mu = k_b T ln(\frac{\Gamma_\uparrow}{\Gamma_\downarrow}) \tag{2.1}$$

(where $\Gamma_\uparrow$ is the pump rate into the cavity by the laser, $\Gamma_\downarrow$ is the rate of loss of photons to the cavity walls, k_b is the Boltzmann constant and T is the temperature of the cavity [10]).

This means that the number of photons with a given energy in the cavity can be modelled as the Bose-Einstein distribution for bosons and this can be used to determine the likelihood of a mode being occupied - taking into account that this is a continuous function and the energy modes are discrete.

The spectral profiles of absorption, $\alpha(\omega)$ and emission, f(ω) are related by the Kennard - Stepanov relation:

$$\frac{f(\omega)}{\alpha(\omega)} \propto e^{\frac{-\hbar\omega}{k_b T}}, \tag{2.2}$$

(where ω denotes the frequency of the photons and $\hbar$ is the reduced planks constant [3]).

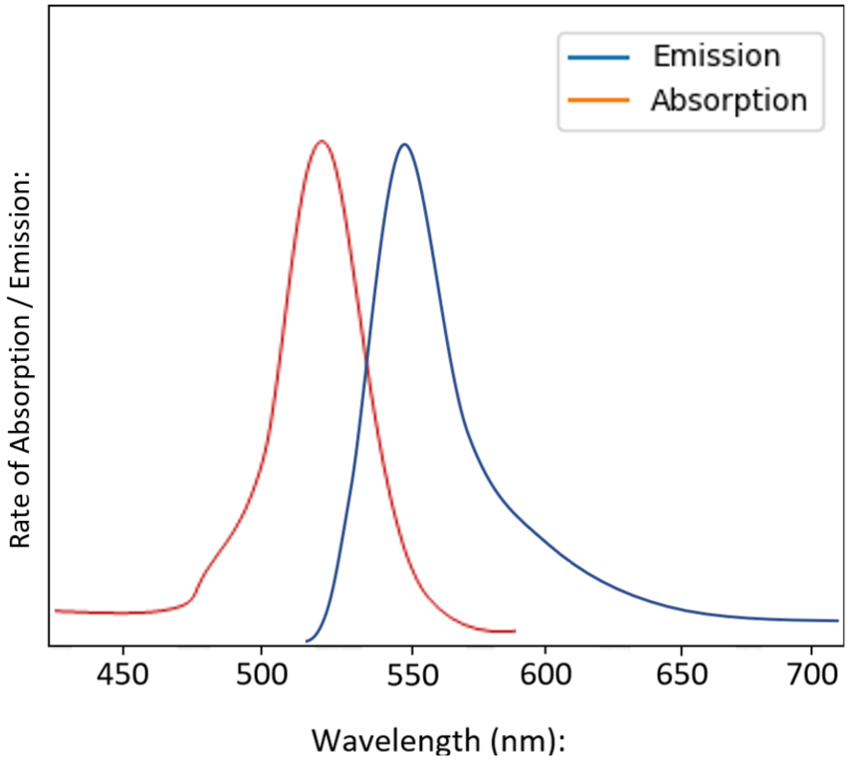


Figure 2.2: Absorption and Emission spectrum of a typical dye molecule - Rhodamine 6G, as a function of wavelength in nm. The red line is of the emission spectrum, and the blue is of the absorption. This figure displays a Stokes shift (difference in peaks of the absorption and emission spectrum) and so the photons emitted by the dye will have lower energy than that which were absorbed. This is called Stokes fluorescence.

Such as in Figure 2.1, the system is pumped with photons at a pump power $\Gamma_\uparrow$. Some of these photons will simply be lost from the system (at a rate $\Gamma_\downarrow$), but some are absorbed and emitted by the dye. This process continues and the photons gradually lose energy - due to stokes fluorescence of the photons from the dye (emission of a longer wavelength photon than absorbed by the dye due to the position of peaks in Figure 2.2). This energy decrease continues until the energies of all the photons reach E_0 and at this point, the system is said to be thermalised. The thermalisation process of photons is significantly different to other Bose-Einstein-Condensate systems since there are no direct interactions between photons in the microcavity.

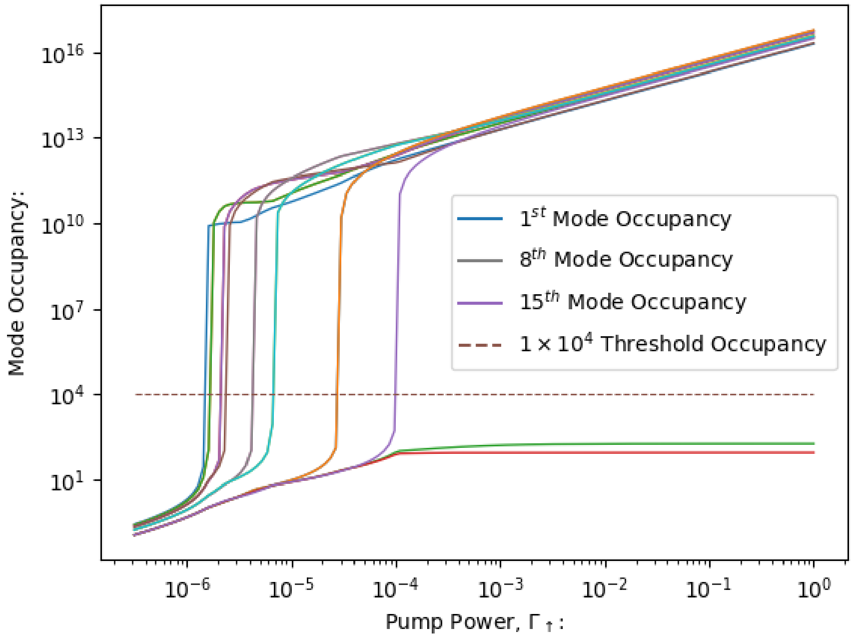


Figure 2.3: The occupancy of the first 15 modes as a function of pump power with a constant cut-off wavelength of the cavity of 598.5nm. This demonstrates the condensing of modes as the pump power increases - due to the large number of photons in the system. It also demonstrates the vast increase in occupancy when a mode condenses. As well as these, it also demonstrates thermalisation breakdown at high pump powers.

The range of cut-off wavelengths used was in the 570-600nm range and as evident in Figure 2.2, both the emission and absorption spectra are decreasing in that range. This means that there are fewer interactions between photons and dye molecules and thus less stokes fluorescence. This means photons are less likely to lose energy to dye molecules and so higher energy modes are occupied - thermalisation breakdown, as evident in Figure 2.3. This means we can define a thermalisation coefficient, γ, which is inversely proportional to one over the cut-off wavelength and which quantifies the amount of thermalisation in the system [7]. The proportionality constant for this relationship is dependent on the type of dye molecule but for this project, but since the type of dye molecule was constant throughout the project, the proportionality constant was assumed to be 1 and thus:

$$\gamma = \frac{1}{\lambda_0}, \tag{2.3}$$

2.2 Neural Networks

A neural network is a computational system which consists of sets of neurons called layers. It has one of these layers as an input layer, one as an output and an arbitrary amount of 'hidden' layers in between. These layers look for particular patterns in the data provided to the neural network to attempt to classify it. This searching occurs using neurons where each neuron is "activated" to a value between 0 and 1 based on the presence of the thing which it is looking. The activation in each neuron can be calculated using:

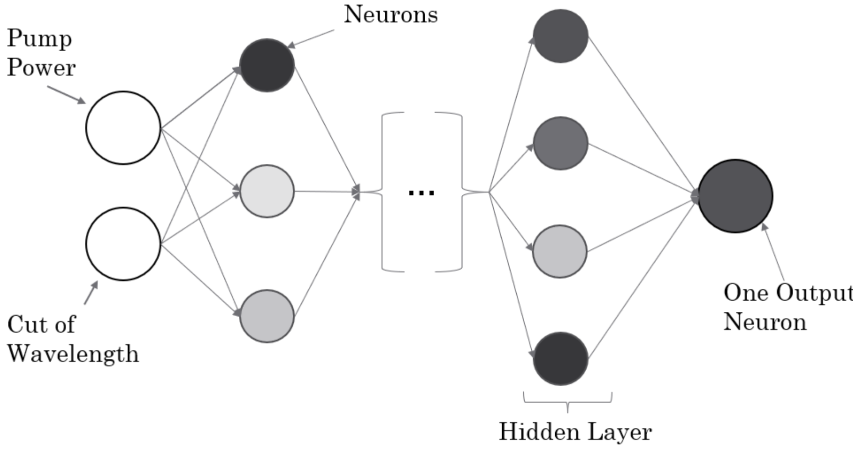


Figure 2.4: A general example of the deep learning neural network used - composed of neurons and layers. It uses two input neurons - the pump power and cut-off wavelength (the parameters of the photon BEC system) and either one output or fifteen output neurons depending on which classification was used (see Chapter 3.1).

$$\mathbf{a^{n+1}} = \sigma(\mathbf{Wa^n} + \mathbf{b}), \tag{2.4}$$

(where $\mathbf{a^n}$ is the matrix of the n^{th} activation's, σ is the sigmoid function, used to scale the calculated values to between 0 and 1, $\mathbf{W}$ is the weight matrix and $\mathbf{b}$ is the bias matrix).

The weight matrix is used to assign the weights of each connecting neuron in Figure 2.4 to the calculation of the activation. Weights are used to assign how much each activation in the previous layer, contributes to the calculation of a particular activation. The bias matrix is used to assign the bias to each activation calculation. Bias is used because it is sometimes required that for a layer to look for particular patterns, some of its neurons only activate for extreme circumstances - which can be achieved by adding a bias to each activation calculation. The bias and weight matrices throughout the neural network are the parameters used to train the neural network to make better predictions and increase its accuracy.

The training of a neural network uses a set of training data. This data is run through the network and for each set of input parameters, it changes the weights and bias' of the network to produce the closest possible output to the given output. This is done by minimisation of the cost function (a measure of the accuracy of a neural networks prediction - often given by the squared difference between the neural networks prediction and the given training output for each neuron, however, different optimisers sometimes use different cost functions) of the neural network using an 'Adam' optimiser (refer to Appendix C) which is run through multiple 'epochs'. For each epoch, the Adam optimiser is run on the cost function to find a new set of weights and bias' for the neural network such that it makes better predictions.

Chapter 3

Method

3.1 Photon BEC Classification

The occupancies of the first 15 modes of the system were measured using a CCD and interferometric techniques (higher modes were not included in the data and their possible inclusion could, therefore, be a potential extension of the project in the future). This was conducted for a range of input pump powers - 0 to 1GHz and cut off wavelengths - 570 - 600nm. These were the two input parameters and the project aimed to find the phase diagram for these two parameters.

To achieve this, it was first determined which of the 15 modes were condensed for each set of input parameters. The criteria for a mode to be condensed was that its occupancy had to be greater than a threshold number. This threshold occupancy has to be in the range of the occupancy of modes which are condensed and uncondensed and using Figure 2.3, values of $1 \times 10^4, 1 \times 10^6$ and 1×10^{10} were thought to be sensible preliminary estimates for this. This gave an output of a list of 0's and 1's for each set of input parameters - 0 for uncondensed and 1 for condensed.

From this, the number of condensed modes for each cut-off wavelength and pump power were summed and normalised to give a number between 0 and 1 (where 1 signified 15 modes were condensed). This classification was called the 'length' classification.

Another method of classification was to label each set of 0's and 1's as that binary number and this classification was called the 'binary' classification. (for example, the number 100000000000000 would signify a BEC due to only the ground state being condensed).

3.2 Machine Learning

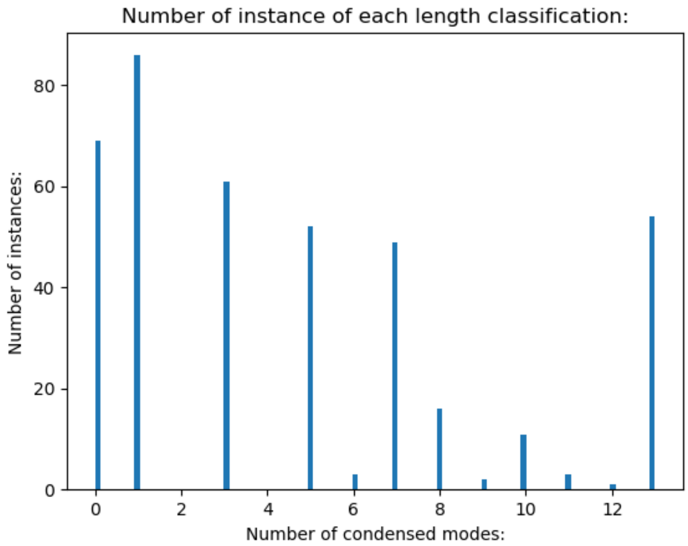


Figure 3.1: Histogram of the number of instances of each length classification for the training data. This was using a 1×10^4 *cut-off occupancy. This histogram demonstrates the data bias and thus the necessity to weight the data for training for each of the different number of condensed modes, with inverse proportion to the number of instances.*

It is vital that the data used to train the neural network has a good data set for each of the outputs - the more the better. This is because the neural network learns from each instance of each data point and so it needs to learn which input parameters return which output. The data was split randomly into test data and train data using random choices. This was done with a 30:70 ratio respectively and was weighted such that each classification was split into test and train data with that ratio. Figure 3.1 shows that despite this weighting in choice, the data was still uneven and so each class was further weighted for training with inverse proportion to the number of instances. This is however still not enough for a length classification of 2 or 4 because there are no instances. This means that the only way to remedy this is to collect more data with these classifications.

A neural network was created with two input neurons - cut-off wavelength, λ_0 and pump power, Γ (for which all λ_0's and Γ's had been scaled to values between 0 and 1). The output layer of the neural network had either 1 neuron corresponding to the scaled number of condensed modes - for length classification, or 15 neurons corresponding to whether each of the 15 modes were condensed or not - for binary classification. When a prediction was taken from a neural network, these output values were denormalised and rounded since a mode can either be condensed or not condensed - a decimal does not make physical sense.

The number of hidden layers and neurons in the neural network was optimised for different threshold values using the length classification. This was first achieved by plotting the mean absolute difference between the predicted number of occupied modes for each parameter combination and the actual data, 'minimum mean error'.

This was repeated for up to 7 hidden layers, each with up to 50 neurons.

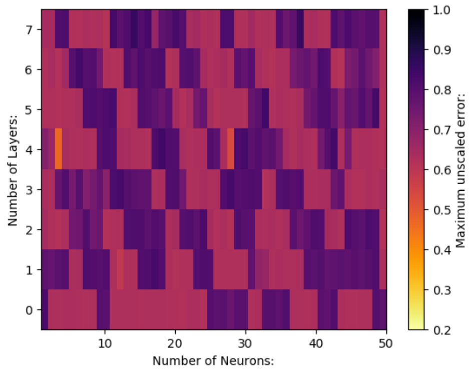
The result of this optimisation was the number of hidden layers and neurons which gave an output with the minimum mean error, was 1 and 29 respectively (the threshold occupancy used had little affect on this result). This seemed unreliable as logically, the more neurons and hidden layers a neural network has, the more training it has undergone and thus the better it should be at learning. The reason this optimisation was unsuccessful was that the mean absolute difference between the real and predicted data, although being a good indicator of absolute error, is a poor judgement of error when comparing neural network outputs since it is a poor method of error for neural networks with few layers and neurons. This is because a neural network with few layers is almost guessing due to its lack of parameters to change (lack of weights and bias'). This means although the mean absolute difference is lower than for a higher number of layers, the spread is much greater. The largest absolute difference between the predicted set of data by the neural network and the set of data was instead used as the error. This optimisation was conducted for three different threshold values to find which combination of neurons, hidden layers and threshold occupancy gave the most accurate neural network.

The binary classification phase plots used the same neural network which was optimal for length (4 hidden layers each with three neurons - except for the output layer having 15 neurons instead of 1). This is because it was assumed that the optimal neural network for length would be approximately optimal for binary (a possible extension to the project could investigate the optimal network for the binary phase plots). However, the threshold occupancy was still investigated for the binary classification. This minimised the error which was defined as the percentage of neural network predictions which were not exactly correct.

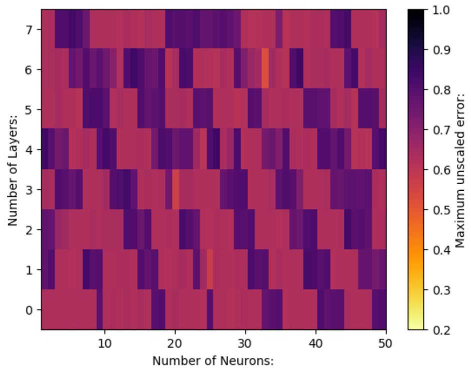
Chapter 4

Results

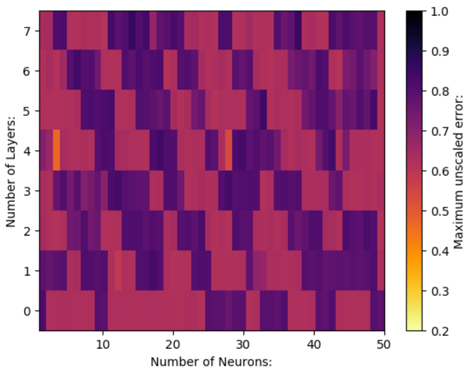
4.1 Neural Network Optimisation



(a) Threshold Occupancy $= 1 \times 10^4$.



(b) Threshold Occupancy $= 1 \times 10^6$.



(c) Threshold Occupancy $= 1 \times 10^{10}$.

Figure 4.1: The maximum unscaled error (difference in the number of condensed neurons divided by 13), between the predicted length classification by the neural network and the actual data set. This is done with neural networks with up to 8 hidden layers and up to 50 neurons. (a) is this with a threshold occupancy of 1×10^4*, (b) is with a threshold occupancy of* 1×10^6 *and (c) is with a threshold occupancy of* 1×10^{10}.

A preliminary investigation of the experiment was first undertaken to maximise the accuracy of the neural network and find the optimal threshold occupancy. Using the length classification, the maximum error between the actual data and the predicted set of data by the neural network was found for three different threshold occupancies (chosen using sensible orders of magnitude from observation of the data, Figure 2.3). This optimisation was completed using different neural networks with up to 7 hidden layers and up to 50 neurons, Figure 4.1.

When the threshold occupancy used was 1×10^4, the errors scaled such as Figure 4.1a. For this network, the optimised number of hidden layers and neurons was 4 and 3 respectively and this combination had a maximum unscaled error of 0.412. When the threshold occupancy was 1×10^6, the errors scaled such as Figure 4.1b. For this network, the optimised number of hidden layers and neurons was 6 and 33 respectively and this combination had a maximum unscaled error of 0.498. When the threshold occupancy was 1×10^{10}, the errors scaled such as Figure 4.1c. For this network, the optimised number of hidden layers and neurons was 4 and 3 respectively and this combination had a maximum unscaled error of 0.453.

This means that the minimum, maximum error is obtained using a threshold occupancy of 1×10^4 in conjunction with a neural network with 4 hidden layers, each of three neurons. This neural network and threshold occupancy was used to obtain the most accurate phase diagram when using the length classification.

This neural network used was quite accurate, but further optimisation methods could be explored. The first of which being that when optimising the neural network, the same number of neurons was used in every hidden layer. Investigation into the maximum error with a variety of neuron number in each of the layers could further improve the precision of the neural network. Another method of improving the neural network is an investigation into the activation of each neuron. This could allow the redundant neurons to be observed and eliminated thus improving the efficiency of the neural network.

4.2 Length Classification

The data was classified using the length classification with a 1×10^4 threshold occupancy. This was split into test and train data and a neural network with 4 hidden layers each with three neurons was used to predict the length phase plot, Figure 4.2. The phase plot was also plotted for the actual data, Figure 4.3.

Verification of the neural network's predictions is that in Figure 4.2, generally, as the number of condensed modes increases, the thermalisation coefficient decreases (which is also true for Figure 4.3). This is what thermalisation is and so makes logical sense.

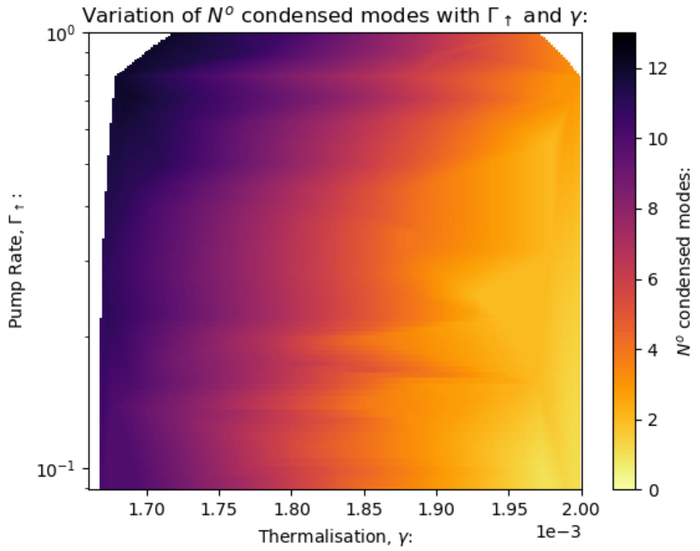


Figure 4.2: Phase plot of a photon BEC as a function of the input parameters of the system, predicted by the optimised neural network. This uses the length classification with a threshold occupancy of 1×10^4. This plot takes discrete predictions and interpolates the values between the data points. The colourmap has missing parts due to missing data provided.

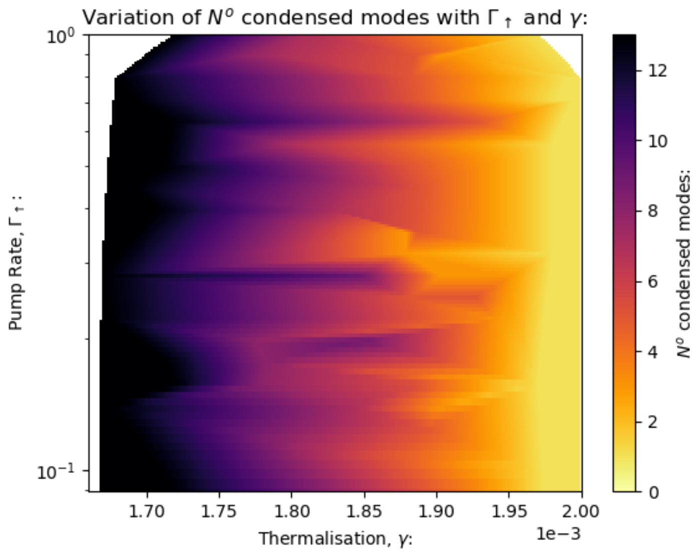


Figure 4.3: Phase plot of a photon BEC as a function of the input parameters of the system, using length classification with a threshold occupancy of 1×10^4. This plot takes discrete predictions and interpolates the values between the data points. The colourmap has missing parts due to missing data provided.

Another verification of the phase plots Figure 5.2 and 5.3 is that the general trend in both figures is the same. This means that the neural network is giving a reliable output.

The final method of verification was to compare the absolute difference between the length predictions made by the neural network and the data:

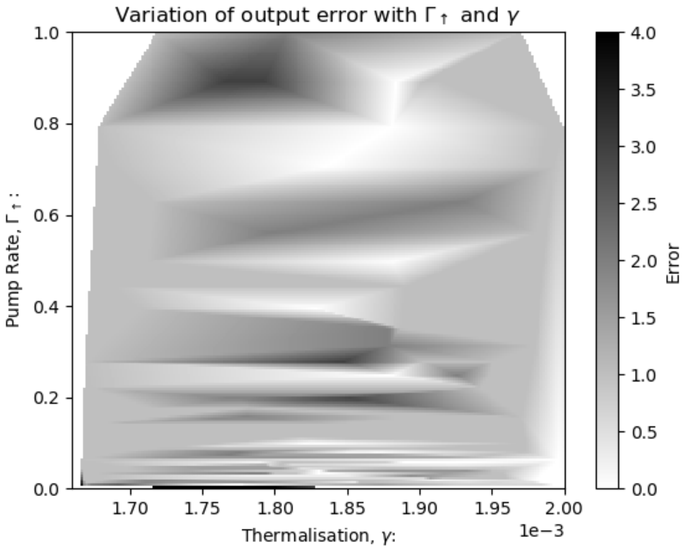


Figure 4.4: Absolute difference in the phase diagram produced by the data and the predicted phase diagram by the neural network. The colormap has missing parts due to missing data provided.

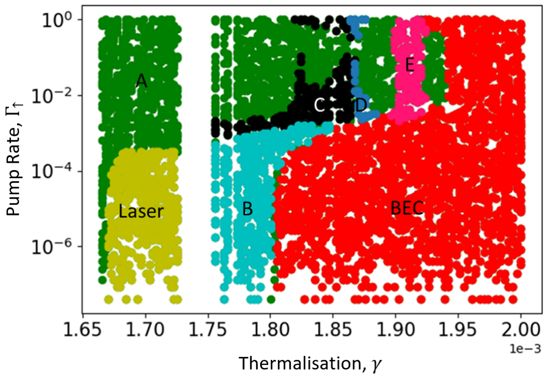
The maximum value of the discrete points from Figure 4.4 was 5.36 (same as the value given when performing the optimisation in Chapter 4.1 if denormalised). This is, however, a poor method of absolute error as it can be very large for just one incorrect data point of many. The mean error in Figure 4.4 was thus calculated instead and used as the absolute error of the length plot = 1.21. This low absolute error demonstrates the accuracy of the neural network used and the reproducibility of the predictions made.

4.3 Binary Classification

The neural network used in the length classification was also used for the binary classification, except with 15 output neurons instead of 1. This was investigated to find the error in the binary classification (the percentage of predictions by the neural network which were not exactly correct) for each of three threshold occupancies - 1×10^4, 1×10^6 and 1×10^{10}.

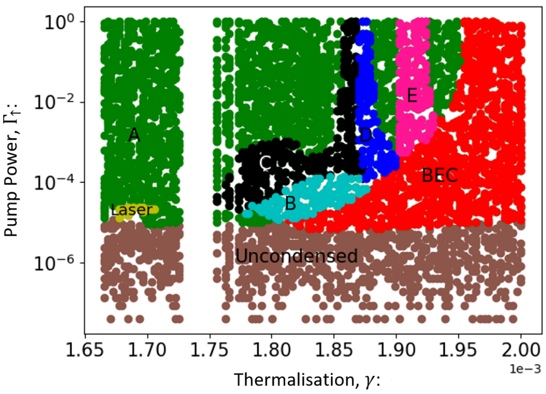
When a binary classification was completed using a threshold occupancy of 1×10^4, 9.51% were found to be not exactly correct whereas, with an occupancy of 1×10^6, 10.8% were found to be not exactly correct. These were compared to when an occupancy of 1×10^{10} was used, which had 13.1% of the predictions not exactly correct.

This shows the optimal threshold occupancy to use was 1×10^4 and so binary phase plots were found using this occupancy:



Label	Binary representation	Condensed modes
BEC	100000000000000	1^{st}
Laser	0xxxxxxxxxxxxxx	Any except 1^{st}
B	111000000000000	$1^{st}, 2^{nd}, 3^{rd}$
C	111110000000000	$1^{st}, 2^{nd}, 3^{rd}, 4^{th}, 5^{th}$
E	100000110000000	$1^{st}, 7^{th}, 8^{th}$
D	100110000000000	$1^{st}, 4^{th}, 5^{th}$
A	Other	Other combinations

Figure 4.5: The predicted phase plot using binary classification by the neural network (with an accuracy of 9.51%*) as a function of the parameters of the system - pump power, $\Gamma_\uparrow$ and the thermalisation coefficient, γ. The threshold occupancy used was 1×10^4. The missing band in the data could emanate from the missing data - as evident in Figure 3.1.*



Label	Binary representation	Condensed modes
BEC	100000000000000	1^{st}
Laser	0xxxxxxxxxxxxxx	Any except 1^{st}
B	111000000000000	$1^{st}, 2^{nd}, 3^{rd}$
C	111110000000000	$1^{st}, 2^{nd}, 3^{rd}, 4^{th}, 5^{th}$
E	100000110000000	$1^{st}, 7^{th}, 8^{th}$
D	100110000000000	$1^{st}, 4^{th}, 5^{th}$
A	Other	Other combinations

Figure 4.6: Phase plot using binary classification as a function of the parameters of the system - pump power, $\Gamma_\uparrow$ and the thermalisation coefficient, γ. The threshold occupancy used was 1×10^4. The missing band in the data could emanate from the missing data - as evident in Figure 3.1.

The main difference in Figures 4.5 and 4.6 is the lack of uncondensed modes in the prediction by the neural network. This problem could stem from the weighting of the classifications in the splitting of the data to rectify the bias in the data. Specifically, it is likely that due to there being an abundance of uncondensed modes in training data, Figure 3.1, these are weighted too far the other way (i.e. they are not weighted enough). Aside from, this the phase plots are relatively similar thus further demonstrating the accuracy of the neural networks used and the reproducibility of the predictions made.

Chapter 5

Conclusion

This work constitutes the phase classification of a photon BEC using a deep learning neural network. This was achieved by observing how the occupancy of different energy modes of a photon BEC changed as input parameters (pump power to the system, $\Gamma_{\uparrow}$ and cut-off wavelength, λ_0 - the wavelength of photons which occupy the ground state energy of both potentials created by the geometry of the mirrors) were changed. To classify the modes, two types of classification were used - length (where the total number of the first 15 modes which were condensed, for input parameters was the phase) and binary (where each mode was denoted a 1 if it were condensed or a 0 if it were uncondensed, giving a 15-bit binary number for the classification of the first 15 modes).

Once the data was classified, it was then split into test and training data with a weighted 30:70 ratio respectively (weighted such that for each classification, 30% of the data was put into test and 70% into training. This data was plotted, Figure 3.1 and was very bias. To counter this, the data was weighted further for training, however, since some classifications had no instances, few improvements could be made.

The neural networks were then optimised to give the most precise predictions to these two types of phase plots. This was first achieved by defining the error as the mean absolute difference between a predicted set of data by the neural network and the test data. It was later discovered that this was a poor method to compare neural network outputs due to the inherent "decisiveness" which a neural network gains as it is trained more (i.e. random guess' - which is close to what neural networks with very few neurons and layers are doing, sometimes give better mean values than those which look for structure in the data). A better method was to define the error as the maximum absolute difference between a predicted set of data by the neural network and the test data. This gave an optimised neural network of 4 hidden layers, each with 3 neurons.

Phase plots were then plotted for the multiple classifications using the optimised neural networks. The length neural network predicted the number of the first 15 modes which were condensed with uncertainty ± 1.21 and the binary neural network predicted the phase of the photon BEC exactly, 90.49% of the time. This demonstrates the degree to which the neural networks can predict the phases and how successful the project was.

There were multiple possible extensions of the project to increase the quality of the results. The first of which was the optimisation of the neural networks, with a varying number of neurons in each layer. This would provide another layer of optimisation and could lead to a greater precision of the neural network. Another possible extension of the optimisation of the neural networks is to calculate the activation's of individual neurons in the network whilst it is being optimised. This would allow the redundant neurons to be eliminated thus improving the efficiency of the neural network. Another possible extension was to find the phase diagrams with the classification of more than 15 energy modes. This could potentially reveal new phases in the phase plots due to more modes condensing. The final possible extension was to re-complete the project using better data. This is because the bias data was thought to be the main source of error in the results.

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Appendix A

BEC Applications

There are many potential applications of BEC research, however, these are extensions on the three main applications of BEC research found, currently being explored:

Quantum Computers require a way to bunch qubits that are all in the same state and a good method of approaching this issue is using BEC's. This is because, as previously mentioned, it consists of a macroscopic number of bosons occupying the same quantum state which is a property which can be utilised when bunching qubits into the same state [4].

Another possible application of BEC research is with quantum decoherence in quantum computers (as the number of bits in a quantum computer increases, quantum decoherence decreases). This is a problem which could be solved by segregating the computing amongst many smaller quantum computers which would work in unison. The individual computers in such a system have potential applications such as the communication of quantum information using BEC's. This enables us to address the decoherence problem by reducing the number of bits necessary for a single computer [5].

And finally, the majority of the detectors with the highest precision currently used come from atom interferometry, using the wavelike properties of atoms to perform interference experiments that measure small shifts induced by these effects. BEC systems may provide an improvement beyond the scope of what is currently achievable with thermal beams of atoms in such experiments [6].

Appendix B

Quantum Harmonic Oscillator

The quantum harmonic oscillator (QHO) is a type of potential of the form Figure B.1.

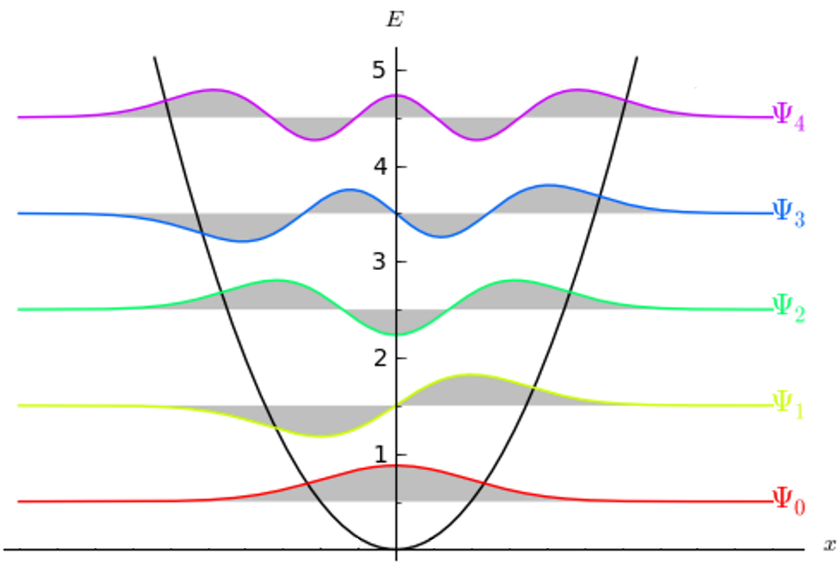

Figure B.1: The form of a one-dimensional quantum harmonic oscillator which can be extended to an arbitrary number of dimensions. It includes the first five energy modes and their wavefunctions. [8]

$$E_i = \frac{\hbar \pi c}{\lambda}(2n_i + 1) \tag{B.1}$$

(where E_i is the energy mode in the i^{th} dimension, $\hbar$ is reduced planks constant, c is the speed of light and λ is the wavelength of the photon which occupies the n^{th} energy mode and n_i is the n^{th} energy mode in the i^{th} dimension).

$$E_{total} = \sum_{i}^{N}(E_i), \tag{B.2}$$

(where E_{total} is the total energy of the particle occupying the system in N dimensions).

Appendix C

Adam Optimiser

The Adam optimiser is an algorithm based on the gradient descent method which minimises a function. In this project, the Adam optimiser was used to minimise the cost function of the neural network by adjusting the weight and bias' of the network. This was achieved with the algorithm:

- Require:
  - α: stepsize.
  - β_1, $\beta_2 \in$ [0,1]: Exponential decay rates for the moment estimates.
  - f(θ): Stochastic objective function with parameters θ.
  - θ_0: Initial parameter vector.
  - m_0, v_0, $t_0 \rightarrow 0$ (initalise 1^{st} and 2^{nd} momentum vectors and the timestep).
- While θ_t not converged:
  - t = t + 1
  - $g_t = \bigtriangledown_\theta f_t(\theta_{t-1})$ - (Get gradients w.r.t. stochastic objective at timestep t).
  - $m_t = \beta_1.m_{t-1} + (1 - \beta_1).g_t$
  - $v_t = \beta_2.v_{t-1} + (1 - \beta_2).g_t^2$
  - $\hat{m}_t = \frac{m_t}{1-\beta_1^t}$
  - $\hat{v}_t = \frac{v_t}{1-\beta_2^t}$
  - $\theta_t = \theta_{t-1} - \frac{\alpha.\hat{m}_t}{\sqrt{\hat{v}_t}+\varepsilon}$
- Return θ_t.

(where θ_t are the parameters of the system -weights and bias' of the neural network, g_t^2 indicates the elementwise square $g_t \odot g_t$. Typically, $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\varepsilon = 10^{-8}$.) For further information on the Adam optimiser refer to [12].

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